

motion to small deviations from steady equilibrium flight. For small deviations from equilibrium, the lateral motion is not coupled to the longitudinal motion and the lateral equations can be solved independently. In this section we examine the stick-fixed free response associated with linearized, lateral, rigid-body motion. Here again, these results provide a reasonable description of lateral aircraft dynamics provided that the aircraft is not undergoing large-amplitude oscillations or rapid maneuvers. We should also keep in mind that the separation of the longitudinal and lateral equations of motion depended on our neglecting the angular momentum of the spinning rotors in the aircraft's engines. In conventional airplanes, such gyroscopic effects can sometimes cause significant yaw-pitch coupling.

Since rigid-body lateral motion has three degrees of freedom (sideslip, roll, and yaw), it is characterized by six eigenvalues. The eigenvalues associated with the linearized lateral motion of a typical airplane contain two distinct real roots, one complex pair, and two identical zero roots. The real and distinct roots correspond to two nonoscillatory modes, one heavily damped motion called the *roll mode* and one lightly damped or divergent motion called the *spiral mode*. The complex pair describes a damped oscillatory motion that is called *Dutch roll*. The remaining two zero eigenvalues correspond to simple rigid-body displacement modes. The roll mode is so heavily damped that it is of no concern to pilots or passengers. As we shall see, the nonoscillatory spiral mode can be divergent even in an aircraft with a high degree of directional stability. However, in most aircraft, the spiral mode is convergent or so slowly divergent that it does not create significant problems for either the pilot or the designer. The Dutch roll mode is an out-of-phase combination of sideslipping, rolling, and yawing oscillations that can be very objectionable. In this section we examine these coupled lateral motions in detail.

The free response of a conventional airplane to small lateral disturbances is obtained from the linearized homogeneous system of lateral equations, with both lateral control inputs set to zero. From Eq. (7.7.6), the homogeneous linearized lateral equations can be written as

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & -t_{xz} & 0 & 0 & 0 \\ 0 & -t_{zx} & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} \Delta\hat{\beta} \\ \Delta\hat{p} \\ \Delta\hat{r} \\ \Delta\hat{\xi}_y \\ \Delta\hat{\phi} \\ \Delta\hat{\psi} \end{Bmatrix} = \begin{bmatrix} R_{y,\beta} & R_{y,\bar{p}} & (R_{y,\bar{r}} - 1) & 0 & R_{gy} \cos \theta_o & 0 \\ R_{\ell,\beta} & R_{\ell,\bar{p}} & R_{\ell,\bar{r}} & 0 & 0 & 0 \\ R_{n,\beta} & R_{n,\bar{p}} & R_{n,\bar{r}} & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & \cos \theta_o \\ 0 & 1 & \tan \theta_o & 0 & 0 & 0 \\ 0 & 0 & \sec \theta_o & 0 & 0 & 0 \end{bmatrix} \begin{Bmatrix} \Delta\beta \\ \Delta\bar{p} \\ \Delta\bar{r} \\ \Delta\xi_y \\ \Delta\phi \\ \Delta\psi \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{Bmatrix} \quad (9.2.1)$$